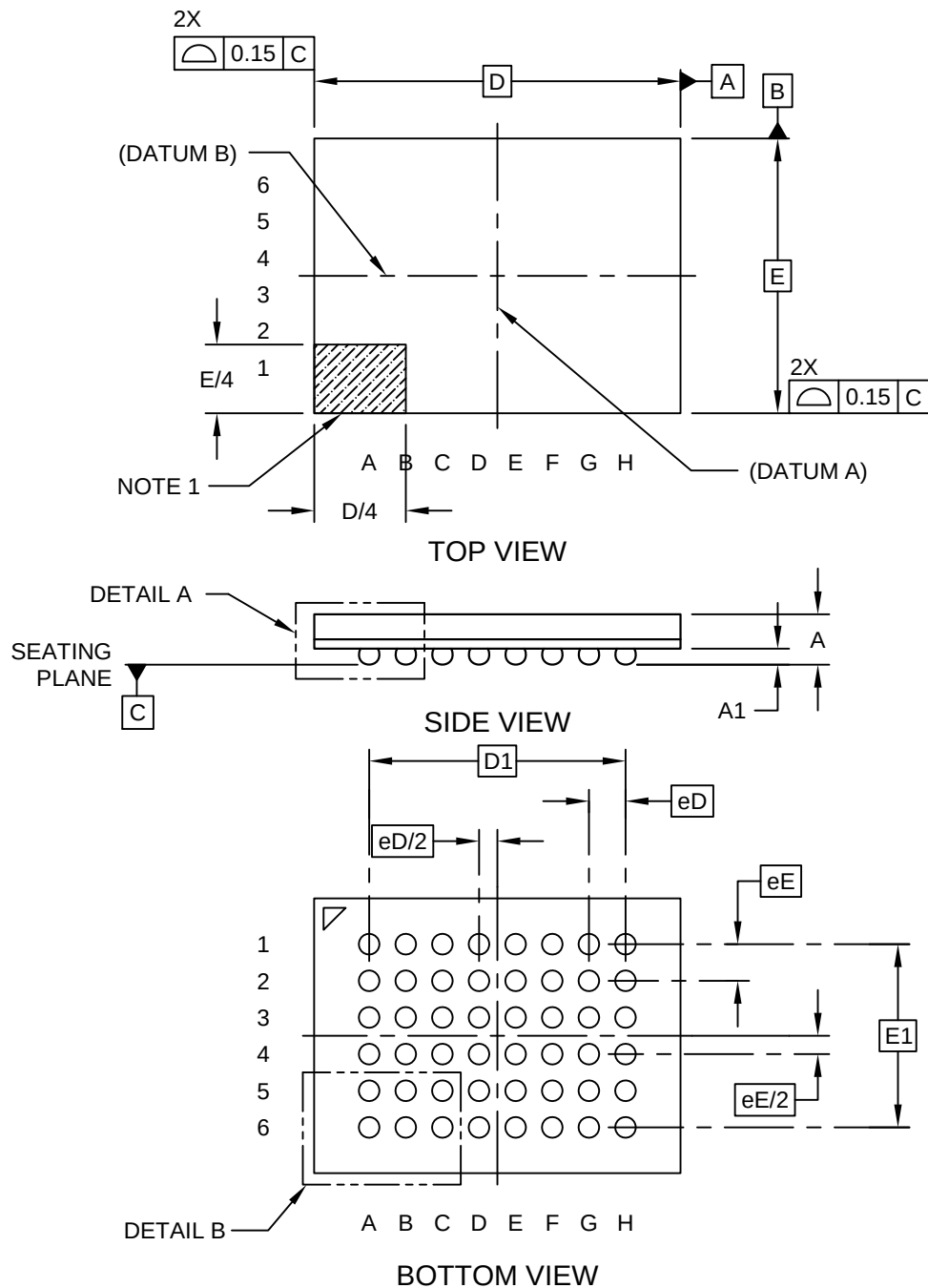


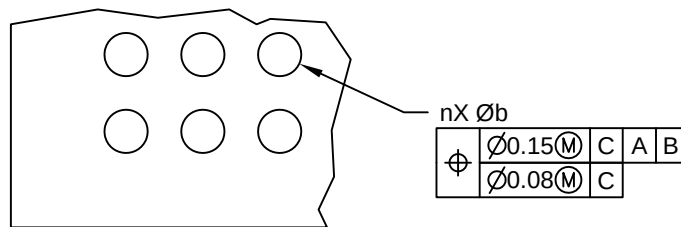
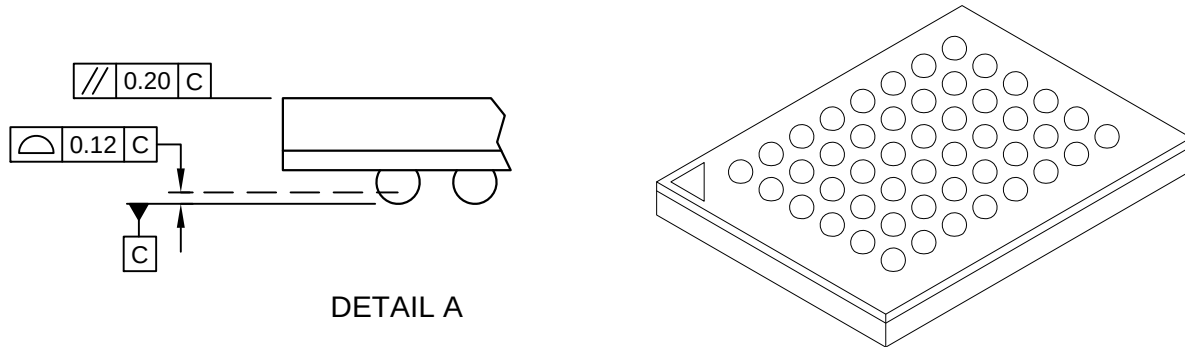
### 48-Ball Thin Profile Fine Pitch Ball Grid Array (CD) - 6x8 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 48-Ball Thin Profile Fine Pitch Ball Grid Array (CD) - 6x8 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



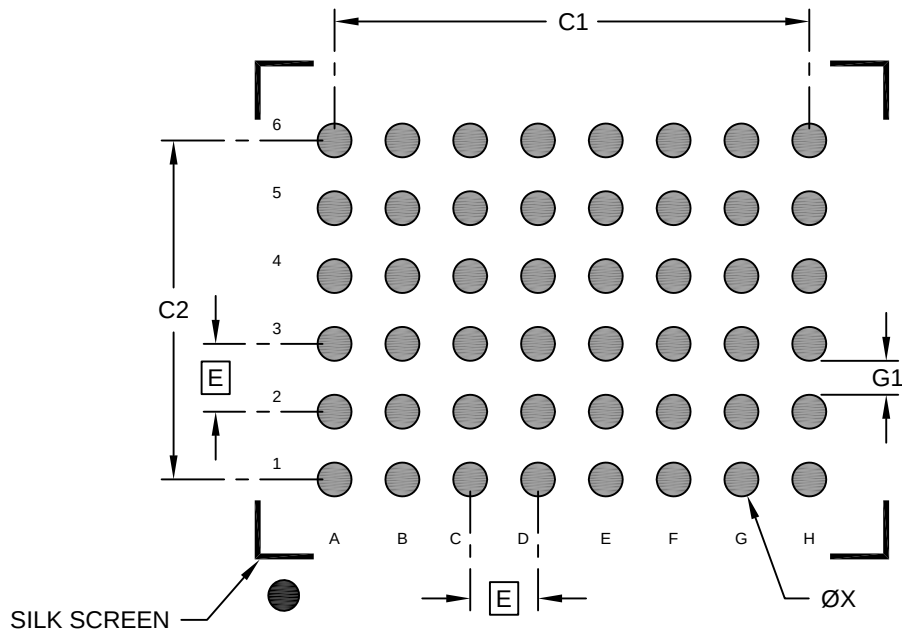
| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Number of Solder Balls      | n  | 48          |      |      |
| Solder Ball X-Pitch         | eD | 0.80 BSC    |      |      |
| Solder Ball Y-Pitch         | eE | 0.80 BSC    |      |      |
| Overall Height              | A  | 1.00        | 1.10 | 1.20 |
| Ball Height                 | A1 | 0.30        | 0.35 | 0.40 |
| Overall Length              | D  | 8.00 BSC    |      |      |
| Overall Solder Ball X-Pitch | D1 | 5.60 BSC    |      |      |
| Overall Width               | E  | 6.00 BSC    |      |      |
| Overall Solder Ball Y-Pitch | E1 | 4.00 BSC    |      |      |
| Solder Ball Diameter        | b  | 0.40        | 0.45 | 0.50 |

### Notes:

- Ball A1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.
- Ball interface to package body: 0.38mm nominal diameter.

## 48-Ball Thin Profile Fine Pitch Ball Grid Array (CD) - 6x8 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                      |    | MILLIMETERS |      |      |
|----------------------------|----|-------------|------|------|
| Dimension Limits           |    | MIN         | NOM  | MAX  |
| Contact Pitch              | E  | 0.80 BSC    |      |      |
| Contact Pad Spacing        | C1 |             | 5.60 |      |
| Contact Pad Spacing        | C2 |             | 4.00 |      |
| Contact Pad Diameter (X48) | X  |             |      | 0.40 |
| Contact Pad to Contact Pad | G1 | 0.40        |      |      |

#### Notes:

1. Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.